10 - Two-Minute Interview Narration

Customer Churn Prediction | Repository-based interview study guide

Source: https://github.com/YuvarajGarla/Customer-churn-prediction

Important: Practice this as a natural explanation, not as a script to recite word-for-word. It deliberately distinguishes repository-reported results from trustworthy validation.

Two-minute narration

The objective of this project was to predict customer churn for a telecom company. Customer churn refers to customers discontinuing their telecom services. Since retaining existing customers is an important business priority, we wanted to identify customers who were likely to leave so that the business could prioritize retention actions.

We used the IBM Telco Customer Churn dataset, which contained around 7,043 customer records and 21 columns. It included demographic information, services subscribed, contract details, tenure, and billing information. The target variable was Churn, and the class distribution was imbalanced, with approximately 73% non-churn and 27% churn customers.

During EDA, we observed higher churn among month-to-month contract customers, customers without online security or technical support, electronic-check users, and customers with shorter tenure.

For preprocessing, we converted TotalCharges to numeric, handled 11 missing observations, created tenure groups, removed the customer identifier and original tenure column, encoded the target, and one-hot encoded categorical variables.

We experimented with Decision Tree and Random Forest classifiers. Since the baseline models had limited recall for churners, the repository explored SMOTEENN, which combines synthetic minority oversampling with nearest-neighbour cleaning. Random Forest with SMOTEENN produced the strongest reported result, at approximately 94% accuracy and 96% churn recall. The notebook also tested PCA, but that configuration performed worse.

One important limitation I identified is that SMOTEENN was applied before the train-test split, introducing data leakage. Therefore, I would not present the resampled test score as a reliable estimate of unseen-data performance. I would improve the experiment by splitting first, applying resampling only within training folds, using stratified cross-validation, tuning the model systematically, and selecting a decision threshold based on business costs.

Shorter 60-second version

I worked on a telecom customer churn prediction problem using the IBM Telco Customer Churn dataset. The goal was to identify customers likely to discontinue service, so the business could prioritize retention efforts. The data contained around 7,043 observations and 21 columns, with an imbalanced target of roughly 73% non-churn and 27% churn. EDA highlighted higher churn among month-to-month contract

customers, customers without online security or technical support, electronic-check users, and customers with shorter tenure. After cleaning, tenure grouping, and one-hot encoding, the repository compared Decision Tree and Random Forest. It explored SMOTEENN to address imbalance and selected Random Forest based on its reported metrics. A critical improvement is to apply SMOTEENN only after splitting, inside the training pipeline, because the original workflow resampled before splitting and therefore leaked information into the test process.

Follow-up questions the narration invites

Q. What was your individual contribution?

Answer honestly based on your actual work. Since this is a repository being studied for interview preparation, do not imply you authored or personally implemented steps you did not perform.

Q. Why did you choose Random Forest?

The repository's reported comparison favoured it after resampling, but a correct leakage-free validation is required to confirm that selection.

Q. What was the biggest issue you found?

SMOTEENN was applied before the train-test split, contaminating the evaluation. The correct approach is to split first and resample training folds only.

Q. What metric matters most?

It depends on the cost of missed churners and false alarms. Recall is important when missing a churner is costly, but precision and campaign cost must also be considered.

Presentation discipline

- Say "the repository reports" when quoting its metrics.
- Do not claim formal hyperparameter tuning or cross-validation; they are not demonstrated.
- Do not claim SMOTEENN results are valid test estimates.
- Be clear about what the original project did versus what you recommend improving.
- Never claim personal implementation ownership unless it is true.